Paper Review • Off-Policy RL / Long-Horizon Value Learning

Horizon-Scalable Q-Learning

A “value-comparison gate” lets each TD target pick its own length

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TL;DR

Extend the TD target a step only if the data action ⩾ the policy

HQL: if the target critic rates the data action ⩾ the best of N behavior-policy candidates, back up through it; else use one-step TD.

→ Auto-selects one- vs. multi-step TD per transition; density-free; fits flow/diffusion policies.

0.74 vs 0.49

Avg. success over 6 OGBench settings
HQL vs. best TD baseline (8-step)

81% vs 62% / 45%

Chunked-HQL avg. success
vs LQL / Q-chunking

0.82 vs 0.53

8 long-horizon OCBench tasks
Chunked-HQL vs 100-step TD

Counterintuitive: 77% of HQL targets stay one-step; it wins on which targets to extend, not how far.

At long horizons, one-step and n-step TD each have a failure mode

1-step TD: residuals add up over horizon

Value error = sum of downstream residuals

$$\varepsilon_t = \Sigma_{\tau \geqslant t} e_\tau$$

Treating residuals as weakly correlated noise:

$$\mathrm{RMSE}(\varepsilon_0) \approx \sqrt{(H \cdot L_{TD})}$$

Small TD loss ≠ accurate value; error grows as $\sqrt{H}$

n-step TD: inherits suboptimal-action bias

Target's deviation from Q^π includes advantages of traversed actions

$$\Sigma_{j=1}^{n-1} A^\pi(s_{t+j}, a_{t+j}) \leqslant 0$$

Same sign, systematic: does not average out

$$\mathrm{Bias} \to H \cdot |\bar{A}_\mu|$$ (grows with n)

Retrace-style corrections need π/μ densities, unavailable for flow/diffusion

Unified error law (Eq. 3)

$$\mathrm{RMSE}(\varepsilon_0)^2 \approx (H/n) \cdot L_{TD}^{(n)} + (1-1/n)^2 \cdot H^2 \cdot \bar{A}_\mu^2$$

Residual accumulation: ↓ as n ↑ Behavior bias: ↑ as n ↑

No fixed n suits the whole trajectory

→ Need a per-transition rule

HQL: a "value-comparison gate" on every data action

Gated multi-step target (Eq. 6)

$$Y_t = r_t + \gamma \cdot V'(s_{t+1}) + \gamma \cdot c_{t+1} \cdot [Y_{t+1} - Q'(s_{t+1}, a_{t+1})]$$

One-step target + gate × downstream correction

$c \equiv 0 \rightarrow$ one-step TD; $c \equiv 1 \rightarrow$ uncorrected multi-step return; Q'/V' are target networks

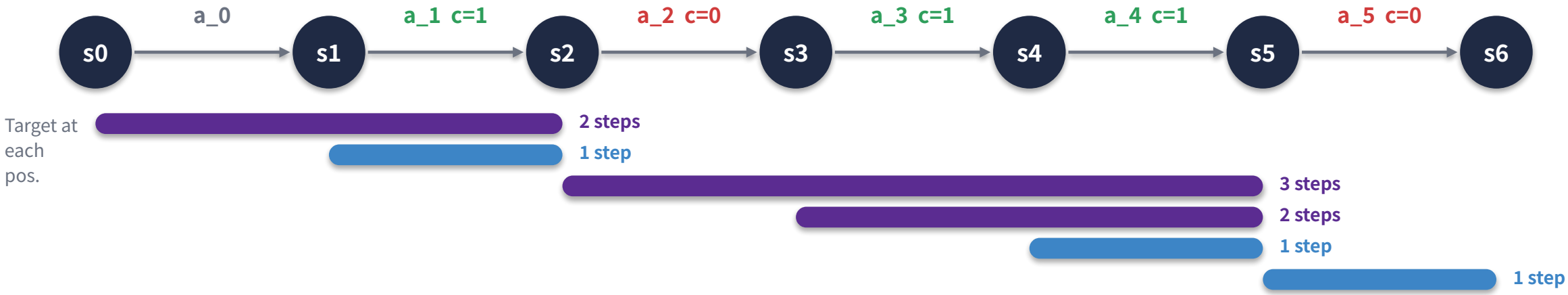
Gate rule (Eq. 7)

$$c_i = \mathbb{I}[\, Q'(s_i, a_i) \geqslant \max_m Q'(s_i, a_i^{(m)}) \,]$$

$a_i^{(m)} \sim \mu(\cdot \mid s_i)$, $m = 1..N$

✓ Ranking only: invariant to reward scale
✓ Density-free: behavior model only samples
✓ Single hyperparameter: candidate count N

Illustration: gates within one window and each target's actual backup length



Source: Section 4.1, Eq. 6–7; diagram below drawn for this review

Hard gate = τ→∞ limit of an importance-sampling "valid gate"

1

Valid gate

$c_i \leqslant \pi(a_i|s_i) / \mu(a_i|s_i)$ preserves the Q^π fixed point (Retrace idea)

2

Exponentially tilted policy

$\pi \propto \mu \cdot \exp(\tau Q) \Rightarrow \log(\pi/\mu) = \tau Q - \log Z$; μ cancels

3

Soft gate (Eq. 9)

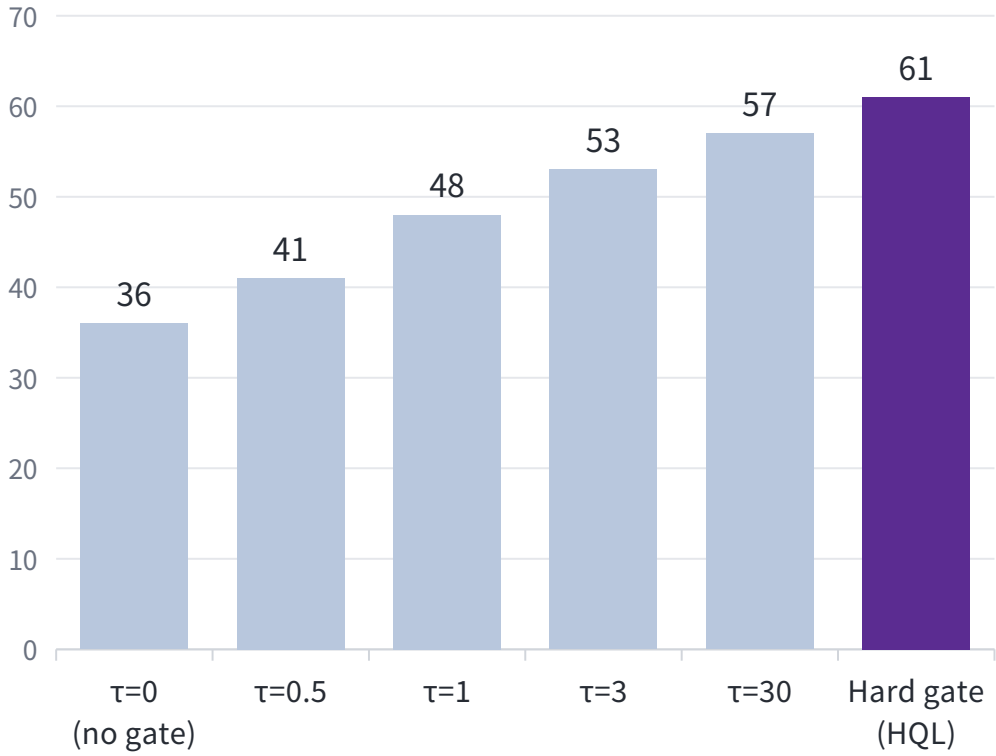
$c_i = \exp(\min\{0, \tau Q'(s_i,a_i) - \log \hat{Z}(s_i)\})$; $\hat{Z}$ via LSE over N samples

4

Two limits

τ→0: gate fully open = uncorrected n-step; τ→∞: exactly the hard comparison (Eq. 7)

Avg. success (%): soft vs. hard gate



Larger τ is better; the hard gate is best: hmaze-medium 91 vs. 77 (best soft gate). Soft gates give suboptimal actions "partial credit", still inheriting some bias.

Source: Section 4.2, Eq. 8–9; right chart from Table 7 (single seed)

One framework covers one-step TD, n-step TD and Q-chunking

Dense targets within a window

Sample H consecutive transitions, compute gates at all positions, then recurse backward from the window end; every (s,a) regresses to its own target.

Fair budget: fixed number of critic targets per update (512 on OGBench); HQL samples B/H windows.

Chunked-HQL

One decision = a chunk of L primitive actions, scored by the critic as Q(s_t, a_{t:t+L}); H chunks span up to L • H environment steps.

Gate unchanged: compare the data chunk with the best-of-N policy chunk; γ becomes γ^L.

L (steps per decision)	H (decisions per target)	Gate	Equivalent method
1	1	—	One-step TD
1	H > 1	No	Uncorrected H-step TD
1	**H > 1**	**Yes**	**HQL**
k	1	—	Q-chunking (chunk length k)
k	**H > 1**	**Yes**	**Chunked-HQL**

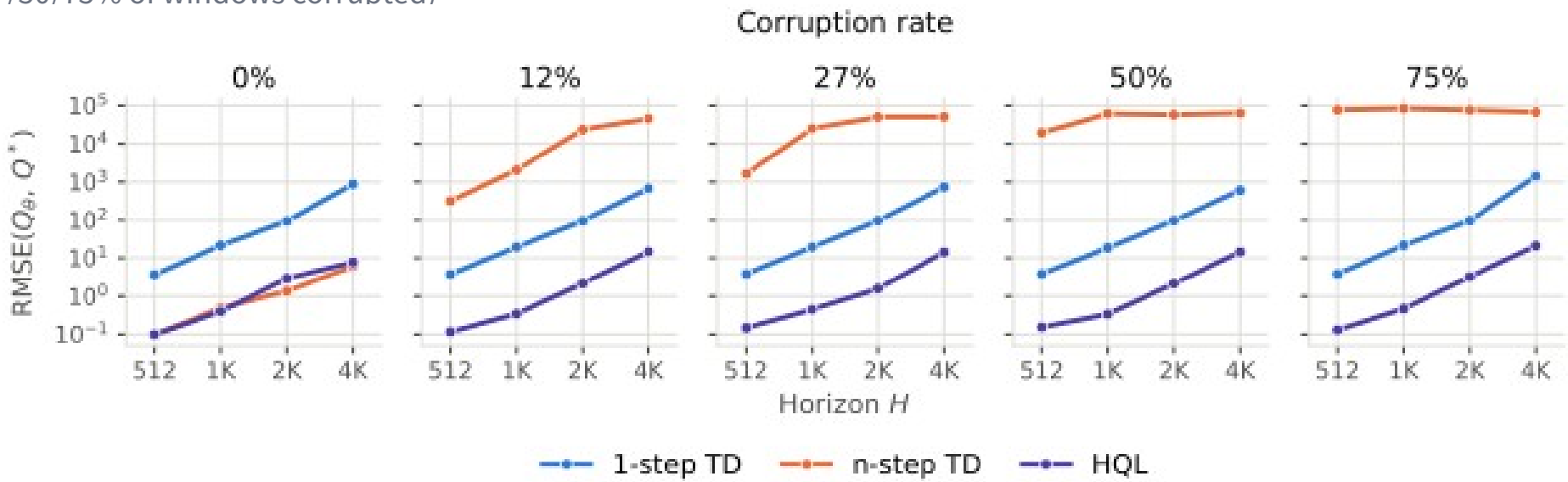
Larger H: a target can chain more decisions; the gate decides how many are used.

Larger L: open-loop execution within chunks; the policy must search an L • dim(A)-dim space.

Source: Section 4.3, Figure 2, Appendix A.4–A.5

Combination lock: only HQL handles long horizons and corruption

Setup: H chained states; correct action advances, wrong one resets; reward −1/step; 64-step windows, later actions swapped for wrong ones w.p. p (0/12/27/50/75% of windows corrupted)



1-step TD

Unaffected by corruption, but error explodes with H: ~3 orders of magnitude from H=512 to 4096

64-step TD

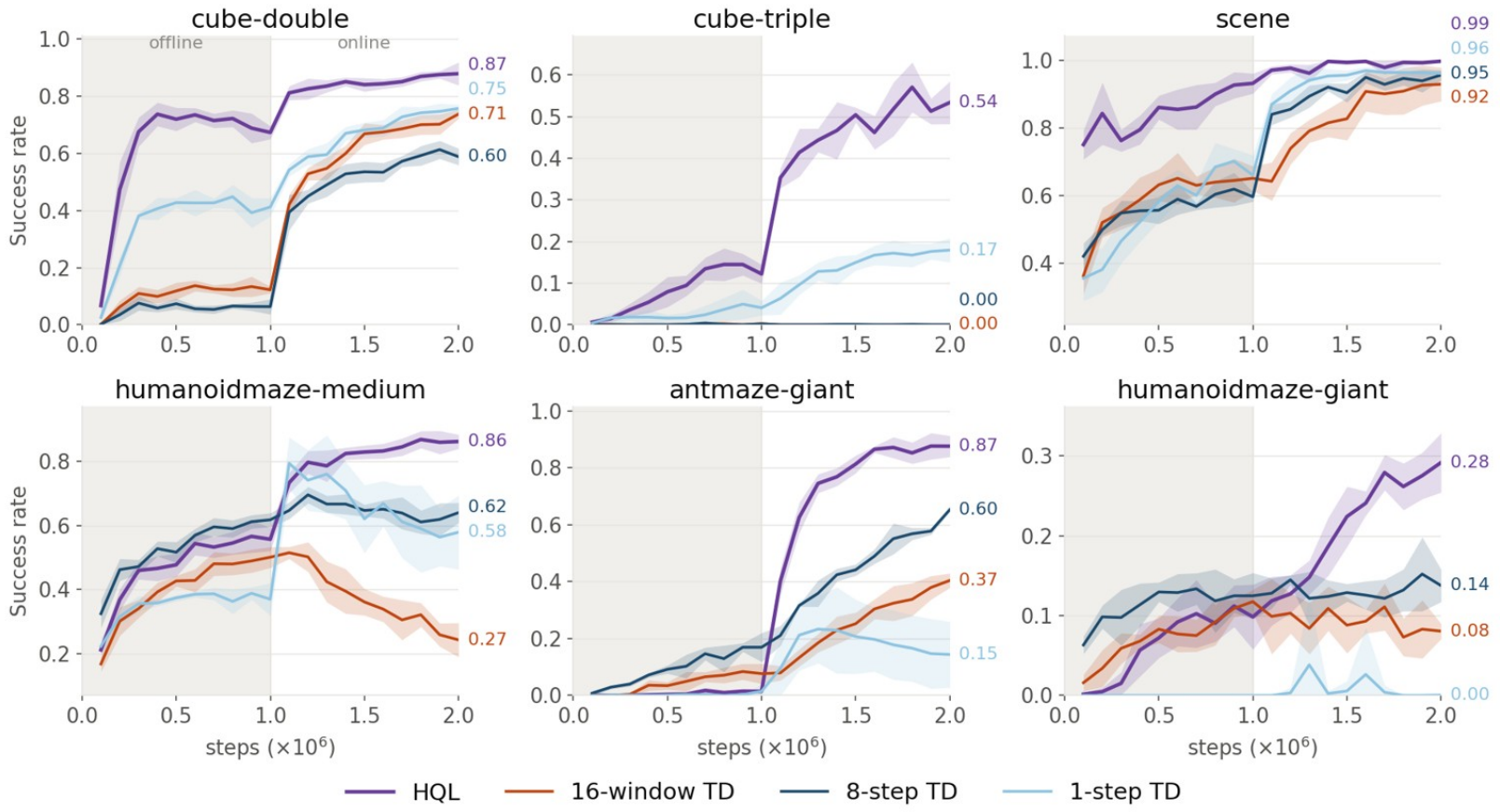
Matches HQL on clean data; diverges under any corruption (RMSE ≈ 10^3–10^5)

HQL

Lowest error at every corruption rate and horizon: truncates backups at corrupted actions

Source: Figure 3 (RMSE, log scale, after 1M updates)

OGBench: HQL beats all TD baselines on all 6 settings



0.74 vs 0.49

Avg. success: HQL vs. best TD baseline (8-step TD)

Largest gap at long horizons

cube-triple: 0.54 vs. 0.00 for multi-step baselines

antmaze-giant： 0.87 vs 0.60

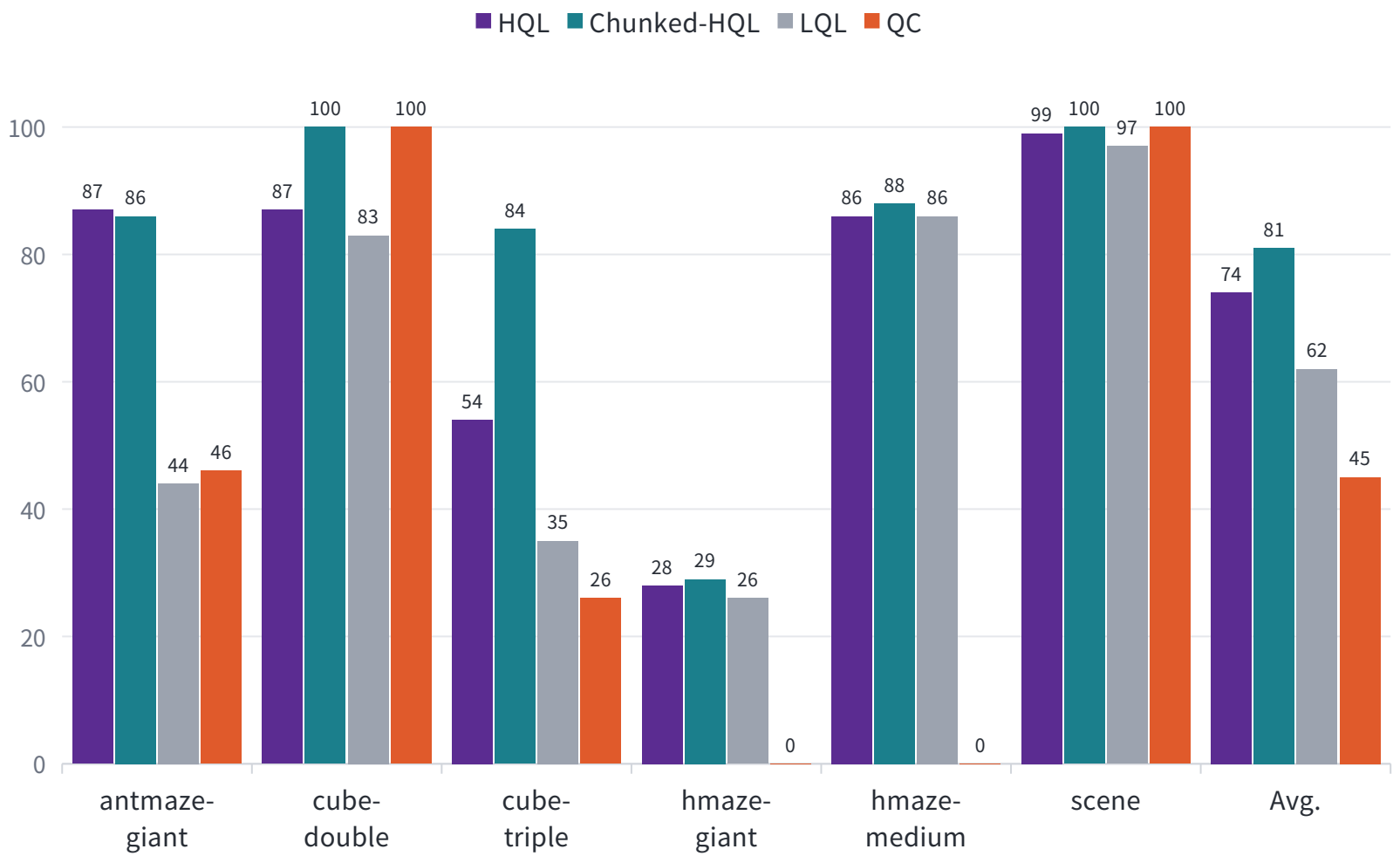
hmaze-giant： 0.28 vs 0.14

The gate is key

16-window TD differs from HQL only in the gate, yet is weakest on average; hmaze-medium drops to 0.27 online

Source: Figure 4 (5 tasks × 5 seeds; gray = offline phase)

vs. LQL/QC: Chunked-HQL avg. 81%; QC gets 0 on humanoidmaze



Why QC fails on humanoidmaze

Action chunks have L・dim(A) dims (21×8); best-of-N coverage is sparse

Open-loop within chunks; no mid-chunk correction

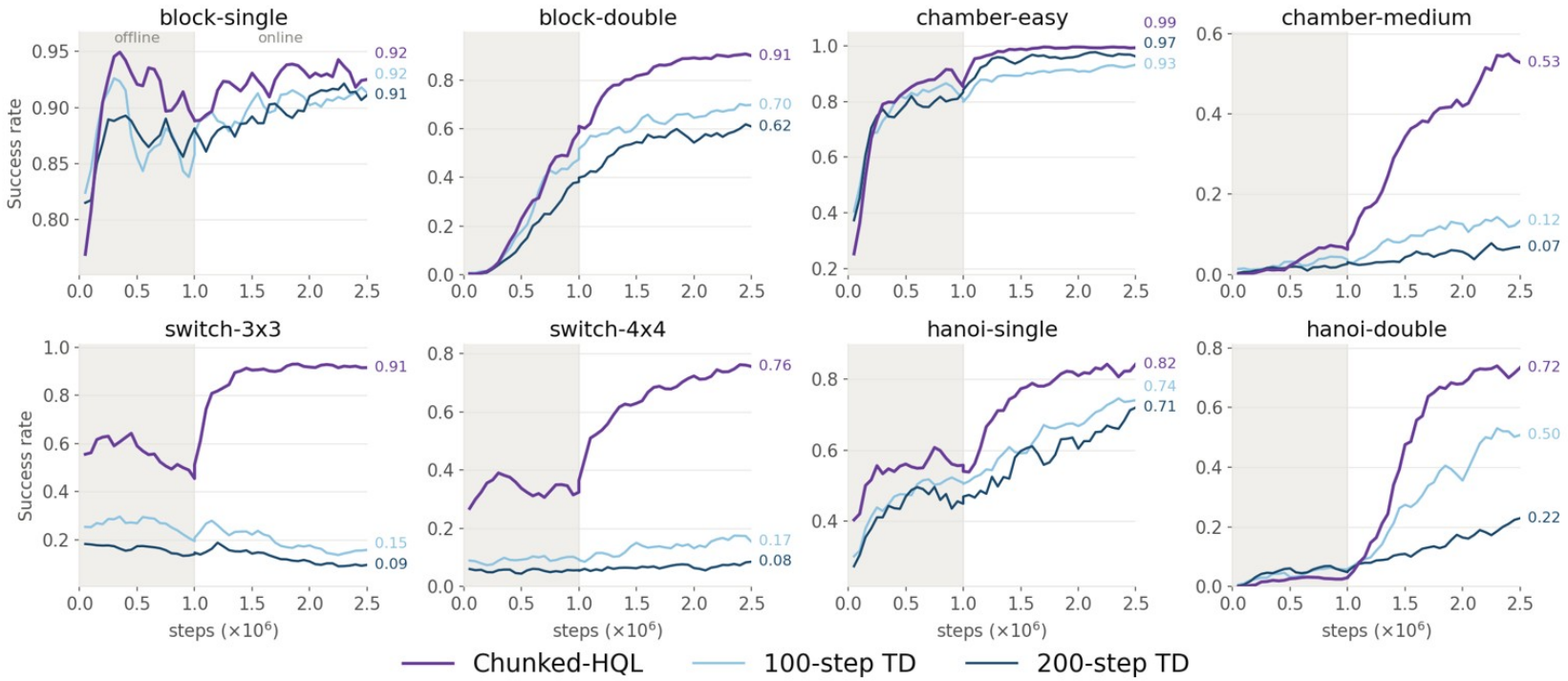
Gate+chunk synergy

45% → 81%

Gate added to QC's action chunks (Chunked-HQL); HQL beats LQL by 12 points on average

Source: Table 1 (success % at end of online training, 5 seeds)

OCBench: Chunked-HQL averages 0.82, far above 100-step TD (0.53)



0.82 vs 0.53 / 0.46

Mean: Chunked-HQL vs. 100 / 200-step TD

Setup

L=25, H=8 (200-step window); gate N=2, execution best-of-16

Largest gap

switch-3x3： 0.91 vs 0.15

switch-4x4： 0.76 vs 0.17

chamber-medium： 0.53 vs 0.12

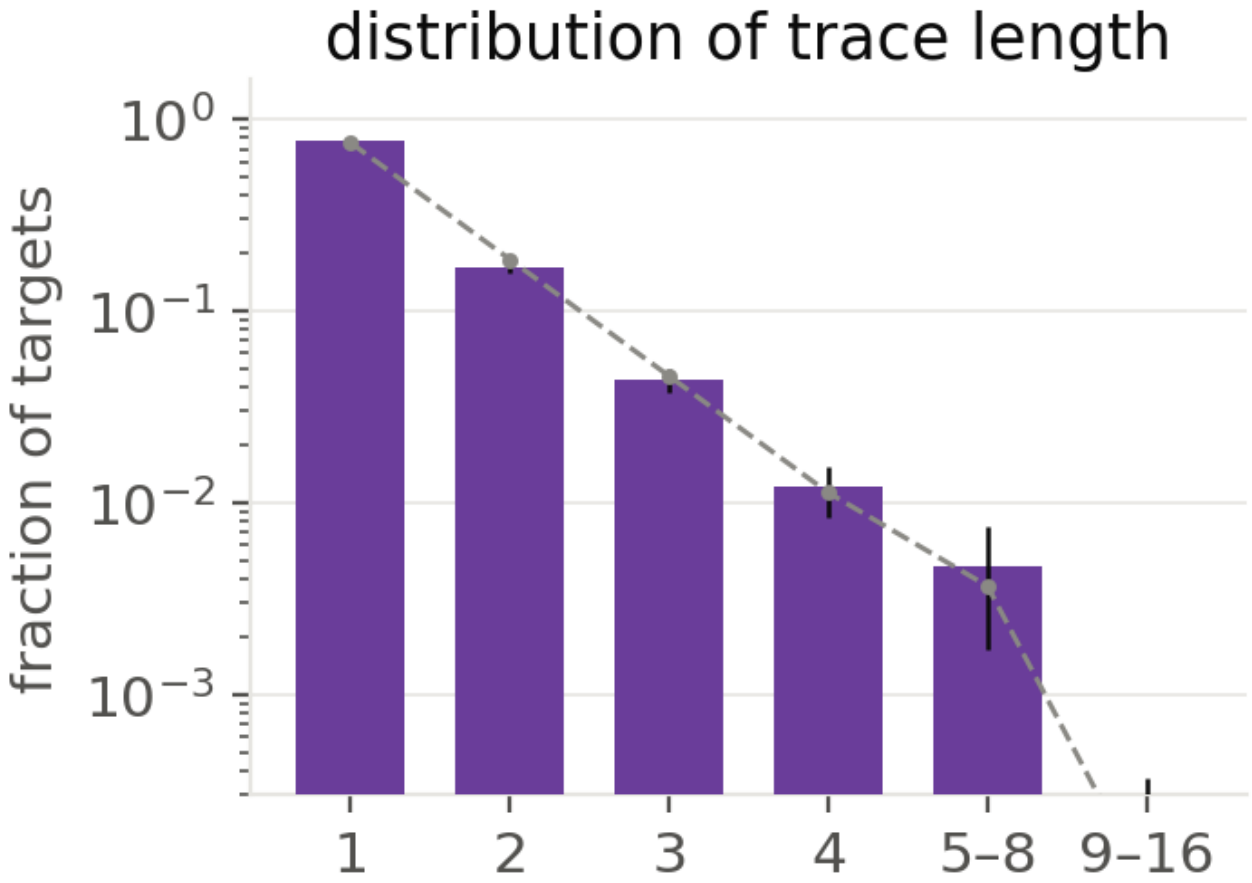
Supports the error law

200-step TD underperforms 100-step on 7/8 tasks

Note: only 1 seed per setting

Source: Figure 5 (OCBench, 8 tasks at 50 Hz; 1 seed per setting)

Surprise: 77% of targets are one-step—HQL wins on which to extend



77% One-step targets

17% Two-step targets

6% Three or more steps

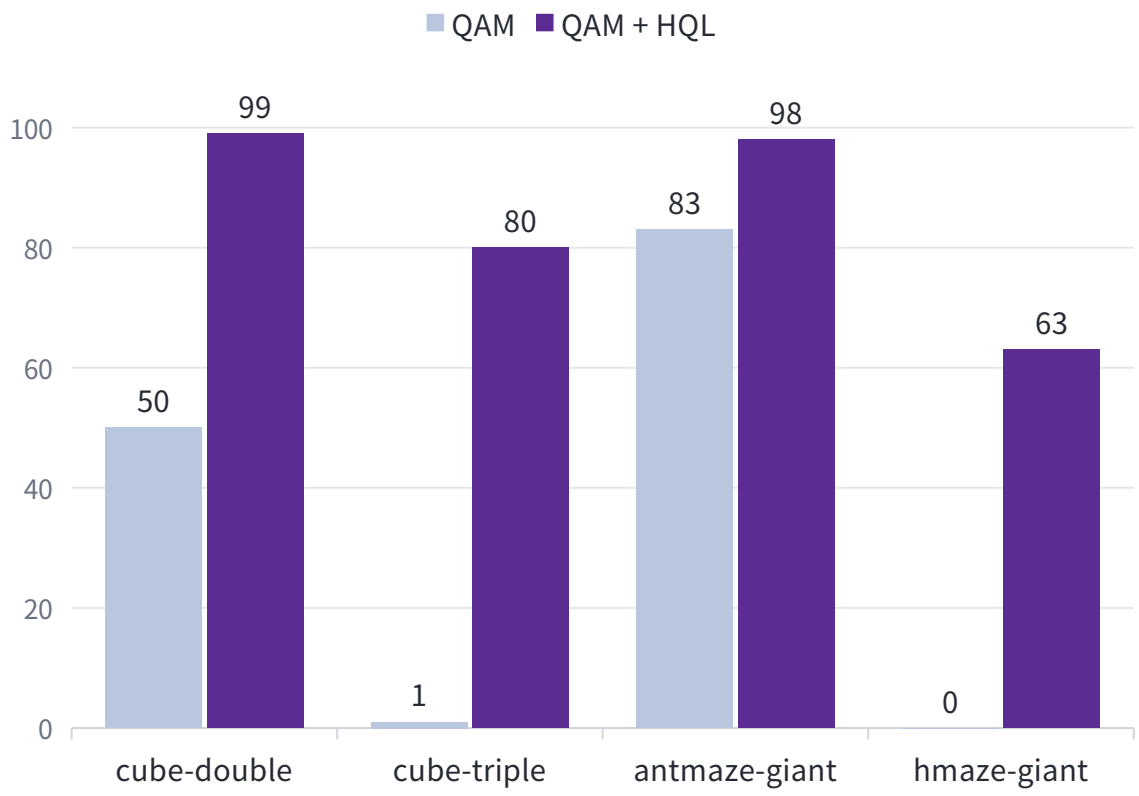
No target exceeds 6 steps (16-step window)

n-step TD extends all → inherits every bias
1-step TD extends none → residuals accumulate
HQL extends only targets that pass the gate

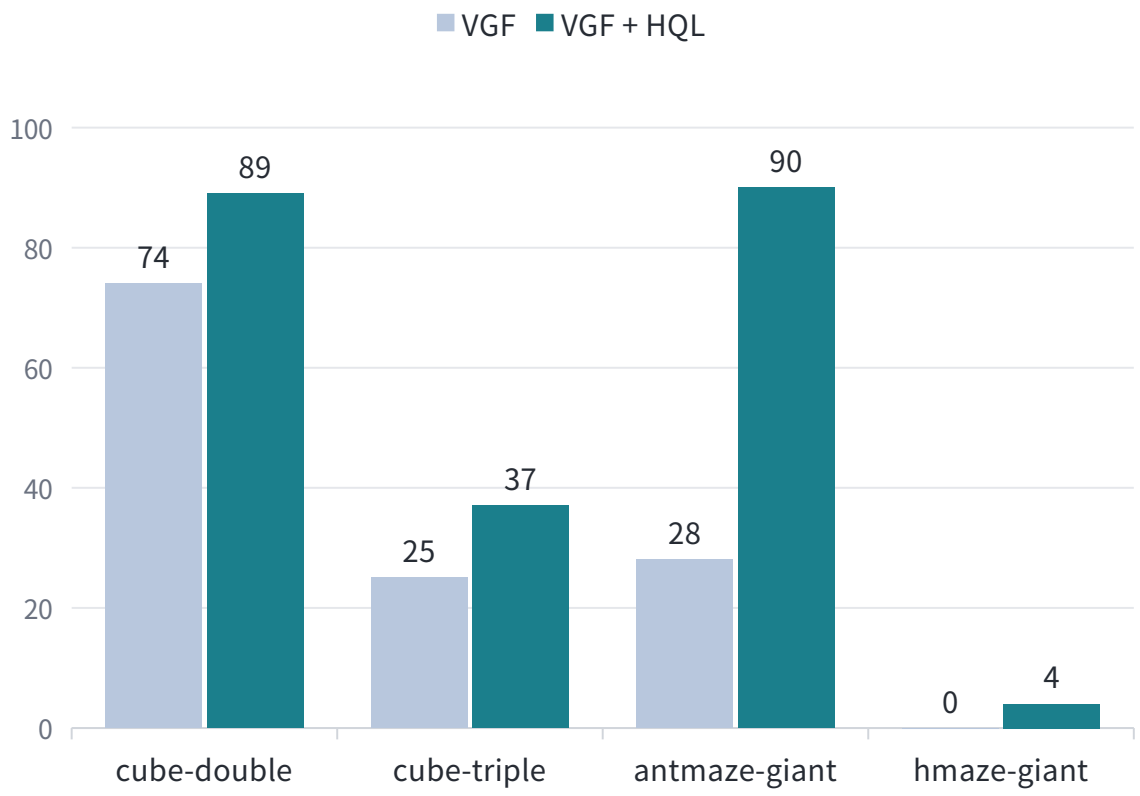
Source: Section 5.3 backup-length distribution (OGBench, window H=16)

Plug-and-play: HQL critics improve QAM and VGF on all four tasks

QAM (adjoint-matching flow policy)



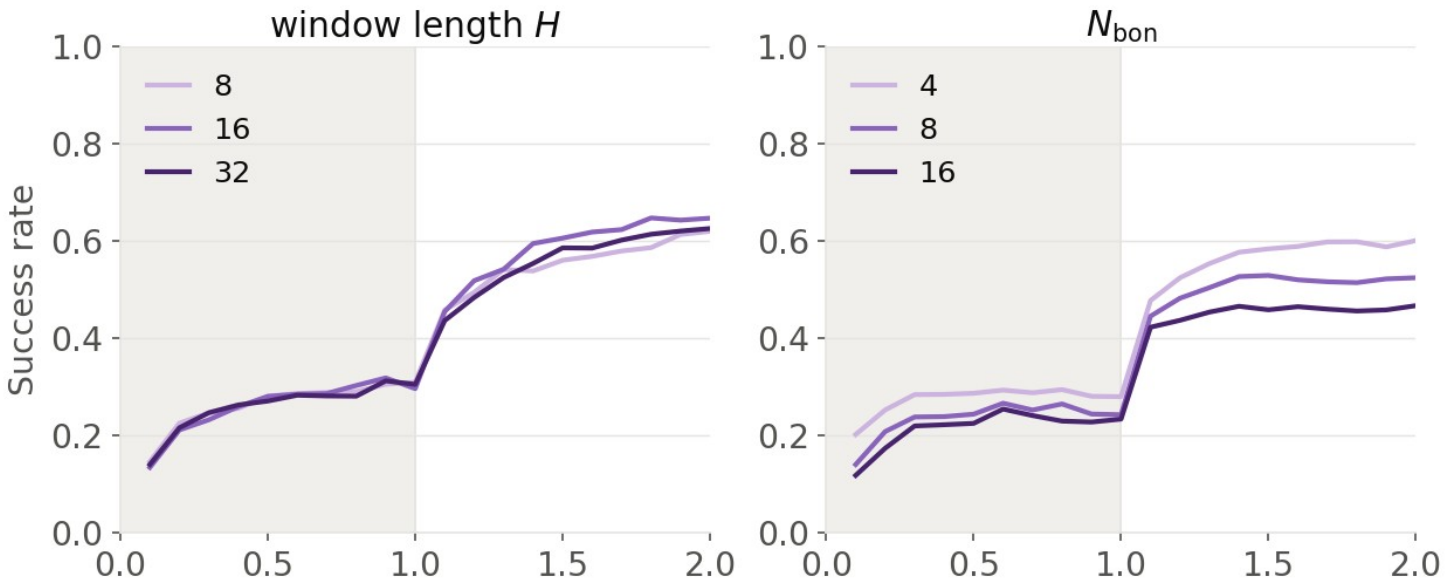
VGF （value gradient flow）



Highlights: QAM cube-triple 1→80, hmaze-giant 0→63; VGF antmaze-giant 28→90 (hmaze-giant only 0→4)

Source: Table 2 (task 2, success % at end of online training)

Ablation: insensitive to window length H; smaller N is better



Window length H

0.607 / 0.646 / 0.621

H = 8 / 16 / 32, largely insensitive: actual targets ⩽ 6 steps, so H=16 suffices

best-of-N width

0.596 / 0.521 / 0.461

N = 4 / 8 / 16; smaller N is better

Data action beats all N candidates w.p. ≈ 1/(N+1): large N → gate rarely opens → reduces to one-step TD

Verdict: simple and principled, but relies on critic's own ranking

Strengths

- Unified error law explains 1-step / n-step TD failures; clear motivation
- Minimal: only changes critic target; sole hyperparameter N; density-free
- Theory: hard gate is the $\tau \to \infty$ limit of the valid IS gate
- General: plug-and-play critic for QAM and VGF
- Fair control: 16-window TD isolates the gate's effect

Limitations & Open Questions

- Gate relies on the critic's own ranking; may wrongly pass/truncate
- Fixed-point guarantee limited to tabular case and exponentially tilted policies
- Conservative: larger N approaches one-step TD; N needs tuning
- OCBench and soft gate: 1 seed only; N ablation: 2 seeds
- Gains mostly online (antmaze-giant offline only 1%)

Source: our analysis, based on Section 6 and experimental setups

Three Takeaways

01

Long-horizon value error has two sources

1-step TD residuals grow as $\sqrt{H}$; n-step behavior bias grows with n; no fixed n works.

02

A value-comparison gate decides step by step

Back up only if data action $\geqslant$ best-of-N; density-free $\tau \to \infty$ limit of the valid IS gate.

03

Wins by choosing which to extend, not how far

77% of targets stay one-step, yet HQL leads widely on long-horizon OGBench/OCBench.

Outlook: apply HQL at the subgoal level for hierarchical long-horizon Q-learning. • arXiv 2611.10987